4.1 Motivation for Sparsity-Aware Learning

Why we want models that can set coefficients to zero.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

The figures follow Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*, and Theodoridis, *Machine Learning: A Bayesian and Optimization Perspective*.

1. Bias-variance trade-off

A simple linear model is

$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_l x_l + \epsilon.$

Plain least squares is not the only way to estimate θ. The goal of this module is better **accuracy** and better **interpretability**.

Regime	What least squares does
$N \gg l$, roughly linear	Low variance, low bias
$N > l$ but not by much	High variance; a small change in the training rows can swing the coefficients; **overfitting**
$l > N$	No unique least-squares estimate; variance is “infinite”

As the model gets more flexible, **interpretability** usually falls. Irrelevant columns add complexity. Setting some coefficients **exactly to zero** would drop those variables.

Solution. Constrain or **shrink** the coefficients: give up a little bias to cut variance, and predictions improve.

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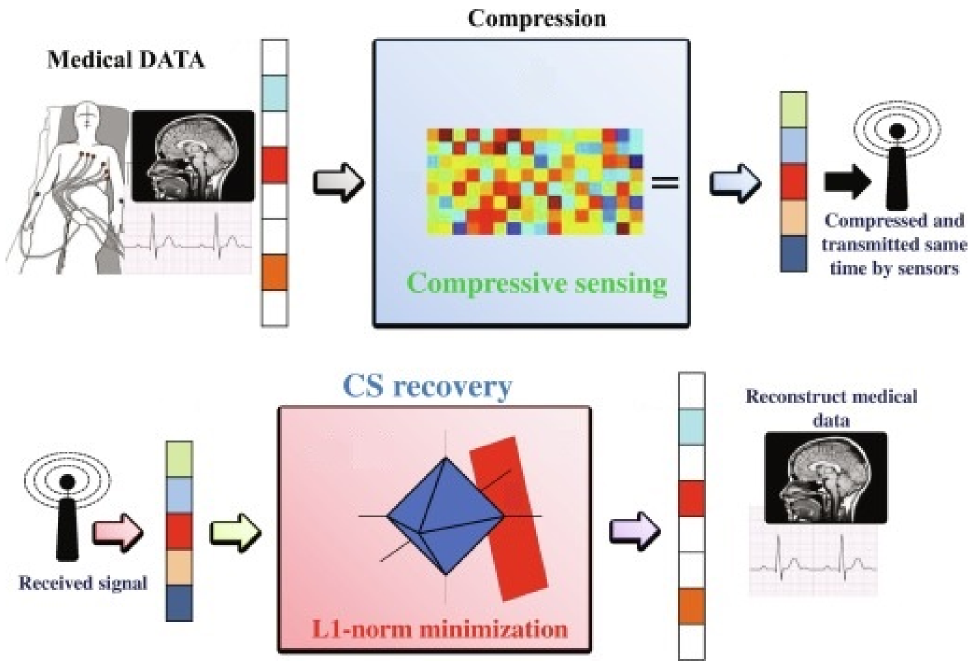
2. Compressed sensing

Compressed sensing (CS) tries to acquire as few samples as possible, as long as those samples still encode a compressed representation of the signal.



Let $\mathbf{X} \in \mathbb{R}^{N \times l}$ be a sensing matrix applied to an unknown signal $s \in \mathbb{R}^{l}$ to get observations $y \in \mathbb{R}^{N}$. Let Ψ be a dictionary in which s has a **sparse representation**

$s = \Psi\theta\,.$

From $y = \mathbf{X}s$ one gets $y = \mathbf{X}\Psi\theta$. Given $\mathbf{X}$ and Ψ (or the product $\mathbf{X}\Psi$), one can recover $\hat{s}$ by an ℓ_1 program if at most k coordinates of θ are nonzero.

The linear system is **under-determined**, so it has an **infinite** number of solutions in general. The extra fact is that the true model is **sparse**: only a few coordinates are nonzero.

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3. The ℓ_0-norm

The ℓ_0-“norm” counts nonzero coordinates. That is the most direct sparsity measure.

Comparing <math xmlns=

ℓp norms" src="graphics/4.1-sparsity-motivation/norms.png" />

It is not usable as a training penalty: it is **not differentiable**, and optimizing it is **NP-hard**. Other ℓ_p with $p < 1$ are **non-convex**, which is also a hard optimization problem. Later notes use ℓ_1 and ℓ_2 instead.

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4. Practical examples

- **NLP.** Most words in a document never appear, or appear once. A bag-of-words vector is sparse: one coordinate per vocabulary word, mostly zeros.

- **Image processing.** Many pixels are background. Keeping only the useful features or regions gives a sparse representation, which is cheaper to store and to compute with.

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5. Emerging applications

- **Medical imaging.** Sparse MRI aims to reconstruct from fewer measurements, which shortens scans.
- **Signal processing.** Compressive sensing for denoising and reconstruction recovers a sparse signal from undersampled measurements.
- **Recommendation.** User–item matrices are sparse. Sparsity-aware methods can still suggest items when the history is thin (the cold-start setting).

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6. Practice

1. In which regime is the least-squares variance “infinite,” and why?
2. Why is the ℓ_0-norm a natural sparsity penalty, and why do we not optimize it directly?
3. What extra information lets compressed sensing pick one solution from an under-determined system?